THE FUTURE WHERE DIGITAL OBJECTS EXIST IN REAL LIFE

Spatial Computing:

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WHY WE STILL DEPEND ON FLAT SCREENS

Today we:

- Watch screens to learn
- Tap flat apps
- See 2D information mostly

Problem:

Humans live in a 3D world, but computers still use 2D screens

Result:

- Hard to understand complex ideas
- Less realistic learning
- Limited interaction

IMAGINE DIGITAL OBJECTS EXISTING IN YOUR REAL ENVIRONMENT

This is where Spatial Computing comes in.

It allows:

- Digital objects in real space
- Interaction with 3D environments
- Blending real + virtual worlds

Spatial Computing:

"UNDERSTANDING THE TECHNOLOGY BEYOUND THE 2D SCREEN"

Spatial Computing is a technology that allows computers to understand and interact with the real-world 3D space

Augmented Reality

(AR)

Virtual Reality

(VR)

Mixed Reality

(MR)

"WHAT TECHNIQUES DO MACHINES USE TO UNDERSTAND THE REAL ENVIRONMENT?"

@ SENSORS

- Cameras sensors
- Depth sensors

@ TRACKING

- Device location
- Ground Tracking

@ RECOGNITION

- Object recognition
- Pattern recognition

Simple Example

- AR apps place virtual objects on your desk

WHY THE WORLD IS MOVING TOWARD SPATIAL COMPUTING

Gartner selected it because:

- People want natural interaction (not screens)
- AR/VR devices are becoming popular
- AI is improving 3D understanding

80%

of people will interact with smart devices daily through Spatial Computing technologies by 2030

(Gartner Emerging Tech Report, 2025 outlook)

"How Technology is Blending into Everyday Experiences"

Education

- 3D classrooms instead of textbooks
- Virtual labs and simulations

Healthcare

- VR surgery training
- 3D organ visualization
- Safe medical practice

Engineering

- 3D building design
- Virtual walkthroughs
- Early error detection

Retail

- Virtual try-before-buy
- AR shopping at home
- Better product visualization

[1] PRIMARY — Springer Nature (Primary Paper)
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Appel, M., Lahn, F., Buytaert, W. and Pebesma, E. (2018) 'Open and scalable analytics of large Earth observation datasets: From scenes to multidimensional arrays using SciDB and GDAL', ISPRS Journal of Photogrammetry and Remote Sensing, 138, pp. 47–56. https://doi.org/10.1016/j.isprsjprs.2018.01.014

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Chen, L-Z., Lin, Z., Wang, Z., Yang, Y. and Cheng, M-M. (2021) 'Spatial information guided convolution for real-time RGBD semantic segmentation', IEEE Transactions on Image Processing, 30, pp. 2313–2324. https://doi.org/10.1109/TIP.2021.3049332

Journal Credibility Evaluation (Primary Paper)

The KN – Journal of Cartography and Geographic Information (Springer Nature) is a highly credible journal indexed in Scopus and Web of Science. It uses a strict peer-review process to ensure quality and accuracy, and has a recognized Impact Factor, showing strong academic influence. This makes it a reliable source for Spatial Computing research.